Saturated Superfluid Vacuum VII-b

Emergent Geometry and the Dissolution of Quantum Gravity

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Abstract

This paper develops gravitational geometry in the Saturated Superfluid Vacuum (SSV) as a macroscopic limit of the same medium that underlies quantum behaviour. The central claim is that if geometry is emergent from the medium, then the conventional programme of quantising gravity is mis-posed: the task is not to quantise the metric but to derive metric behaviour as a coarse-grained description of the underlying condensate.

The emergent metric follows from the density-to-potential identification $\Phi = b \ln(\rho/\rho_0)$ of Paper IV, with time dilation $d\tau/dt = \sqrt{1 + 2\Phi/c^2}$ reproducing the Schwarzschild line element at post-Newtonian order. Spatial curvature in the full isotropic metric resolves the factor-of-two discrepancy in horizon radius identified in Papers IV and V, yielding the correct surface gravity $\kappa = c^4/(4GM)$ and Hawking temperature $T_H = \hbar c^3/(8\pi GMk_B)$. The full nonlinear Einstein equations are recovered via the Jacobson acoustic-thermodynamic route, with Newton's constant G expressed in terms of the healing length ξ already fixed by the particle sector. At the vortex-core scale $\xi \sim \ell_P$ the continuum description breaks down, and the quantum-gravity problem is replaced by the problem of deriving the continuum limit from the medium.

Contents

1 Introduction

Papers I–IV establish the SSV medium, its topological defects, thermodynamic arrow, and gravitational limit. Paper V shows that the same condensate supports black-hole horizons as acoustic-freezing surfaces. Paper VII-a derives the quantum-mechanical limit from the Madelung transformation. The present paper closes the geometry side: it asks whether the metric itself is emergent, and if so, what becomes of the standard quantum-gravity problem.

The standard problem arises from trying to reconcile two fundamentally different languages: smooth Riemannian geometry on one side and the discrete probabilistic structure of quantum field theory on the other. In SSV both languages are effective descriptions of the same medium at different scales. The metric is not a primitive structure to be quantised; it is a bookkeeping language for how the medium stores density gradients, sound-speed variation, and irreversible time flow. Quantising the metric directly is then analogous to quantising the wake pattern instead of the fluid that makes it.

Scope

This paper covers: the emergent metric picture, weak-field and post-Newtonian correspondence, the spatial-curvature correction that resolves the factor-of-two gap from Papers IV and V, and the full Einstein equations from the Jacobson [9] acoustic-thermodynamic route. Out of scope: the full QM derivation (Paper VII-a [6]), black-hole ontology (Paper V [5]), and cosmogony (Paper VIII).

Roadmap

Section 2 argues that quantising the metric is the wrong level of description. Section 3 derives the emergent metric from the density field. Section 4 establishes weak-field correspondence and resolves the factor-of-two horizon discrepancy. Section 5 recovers the full Einstein equations and discusses the cosmological constant. Section 6 discusses open problems and position in the series.

2 Why Quantizing the Metric Is the Wrong Level

Gravity and quantum mechanics share a substrate in SSV: both are properties of the same order parameter. Gravity is the long-range density gradient of the condensate (Paper IV [4]); quantum behaviour is the wave mechanics of that condensate in the low-amplitude limit (Paper VII-a [6]). There is no fundamental tension between them because they operate in different regimes of the same equation of state.

The conventional quantum-gravity problem is therefore not a conflict waiting to be resolved at some unification energy. It is a category error: it attempts to impose quantum structure on an emergent description. A metric $g_{\mu\nu}$ is a summary of how the medium's density, propagation speed, and clock rate vary from point to point. Asking "how do you quantise $g_{\mu\nu}$?" is as well-posed as asking "how do you quantise the speed of sound in a fluid?" The answer is: you derive it from the microscopic theory, where quantisation is already present through the order parameter and its topological defects.

The dissolution argument

In SSV the quantum-gravity problem is replaced by a derivation problem: show that the Madelung equations of the condensate, coarse-grained over scales $\gg \xi$, produce metric behaviour consistent with general relativity. Once that derivation is complete, no separate quantisation of the metric is required. The Planck scale is the scale at which the coarse-graining breaks down—the vortex-core granularity of the medium—not the scale at which a new theory must be postulated.

3 Emergent Metric Picture

3.1 Density as gravitational potential

Paper IV [4] identifies the update availability

$$A(\mathbf{x}) \;=\; \frac{d\tau}{dt} \;=\; 1 + \frac{\Phi(\mathbf{x})}{c^2}, \qquad \Phi(\mathbf{x}) = b \ln\!\left(\frac{\rho(\mathbf{x})}{\rho_0}\right), \tag{1}$$

where b is a medium constant with dimensions of velocity^2 and ρ_0 is the unperturbed condensate density. In a weak gravitational field produced by a mass M, matching to the Newtonian potential gives $\Phi \approx -GM/r$ and

$$A(r) \;=\; 1 - \frac{GM}{rc^2}. \tag{2}$$

3.2 Time dilation and the Schwarzschild line element

The local clock rate (1) reproduces the Schwarzschild time-dilation factor at post-Newtonian order. The proper-time interval for a static observer at r is

$$d\tau^2 = A(r)^2\, dt^2 \;\approx\; \left(1 - \frac{2GM}{rc^2}\right) dt^2, \tag{3}$$

where the second equality uses the Schwarzschild result $A_{\mathrm{Sch}}(r) = \sqrt{1 - 2GM/(rc^2)}$.

3.3 Emergent metric tensor

Writing the Madelung equations [6] in a curved-background notation introduces an effective metric. In the isotropic post-Newtonian gauge, to leading order in Φ/c^2:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad h_{00} = -\frac{2\Phi}{c^2}, \quad h_{0i} = -\frac{2v_i}{c}, \quad h_{ij} = -\frac{2\Phi}{c^2}\,\delta_{ij}, \tag{4}$$

where v_i is the background flow velocity and Φ is the gravitational potential (1). The spatial part $h_{ij} = -(2\Phi/c^2)\delta_{ij}$ is the isotropic-gauge form of Schwarzschild: the metric *is* the density field, with no separate geometric input.

Emergent metric

The effective metric (4) is not imposed on the medium. It is derived from the density perturbation $\Phi = b\ln(\rho/\rho_0)$ and the irrotational flow v_i. Time dilation, light deflection, and gravitational redshift are properties of the medium's equation of state, not postulates of a separate geometric theory.

4 Weak-Field Correspondence

4.1 Newton's constant from acoustic coupling

Paper II [2] shows that two breather-mode defects (protons, or any oscillating density perturbations) at separation r experience an attractive Bjerknes acoustic force [7]:

$$F_{\mathrm{Bj}} = -\frac{\partial}{\partial r}\langle p_1 V_2 + p_2 V_1\rangle \approx -\frac{G_{\mathrm{eff}}\, m_1 m_2}{r^2}, \tag{5}$$

with effective coupling constant

$$G_{\text{eff}} = \frac{\omega_0^2 A_0^2}{4\pi\rho_0 c_s^2}, \tag{6}$$

where ω_0 and A_0 are the characteristic frequency and amplitude of the breather modes. Matching $G_{\text{eff}} = G$ (Newton's constant) fixes the combination $\omega_0 A_0/(c_s\sqrt{\rho_0})$ in terms of known constants.

4.2 Post-Newtonian expansion

A gradient expansion of the condensate Euler equation around the saturated ground state $(\rho_0, \mathbf{v} = 0)$ confirms the metric identification (4) at second order in Φ/c^2. At first post-Newtonian order, the stress-energy divergence obeys

$$G_{\mu\nu} \equiv R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{7}$$

where $R_{\mu\nu}$ is the Ricci tensor of the metric (4). This confirms that the acoustic dynamics reproduces general relativity in the weak-field slow-motion limit.

4.3 Spatial curvature and the factor-of-two horizon correction

Papers IV and V flag a factor-of-two discrepancy between the SSV Newtonian horizon radius and the Schwarzschild result. The root cause is the spatial component h_{ij} in (4).

The Newtonian limit (Paper IV) uses $A(r) = 1 + \Phi/c^2 = 1 - GM/(rc^2)$. Setting $A(r_H) = 0$ gives $r_H^{(\text{Newt})} = GM/c^2$, half the Schwarzschild radius $r_S = 2GM/c^2$.

In the isotropic metric (4) the spatial part contributes equally to radial geodesics: the effective potential governing radial null rays picks up both h_{00} and h_{rr}. At post-Newtonian order the result is $g_{00} = 1 - 2GM/(rc^2)$, which gives

$$A_{\text{Sch}}(r) = 1 - \frac{r_S}{r}, \qquad r_S = \frac{2GM}{c^2}, \tag{8}$$

and the horizon condition $A_{\text{Sch}}(r_H) = 0$ correctly yields $r_H = r_S = 2GM/c^2$.

The surface gravity is

$$\kappa = \frac{c^2}{2} \left.\frac{dA_{\text{Sch}}}{dr}\right|_{r_H} = \frac{c^4}{4GM}, \tag{9}$$

and the Hawking temperature from the acoustic Unruh effect (Paper V [5]) is

$$T_H = \frac{\hbar\kappa}{2\pi k_B c} = \frac{\hbar c^3}{8\pi G M k_B}, \tag{10}$$

the standard Bekenstein–Hawking result.

Factor-of-two resolution

The factor-of-two discrepancy in horizon radius (Papers IV and V) and the resulting factor-of-four discrepancy in T_H are resolved by including the spatial curvature h_{ij} of the isotropic metric (4). The Newtonian limit of Paper IV uses only h_{00}; the full post-Newtonian metric includes $h_{ij} = h_{00}\delta_{ij}$, doubling the effective gravitational potential for radial motion and yielding the correct $r_H = 2GM/c^2$, $\kappa = c^4/(4GM)$, and $T_H = \hbar c^3/(8\pi G M k_B)$.

5 Relation to General Relativity

5.1 Jacobson acoustic-thermodynamic route

The post-Newtonian expansion of the previous section confirms the Einstein equations perturbatively. The following argument recovers them nonlinearly via the Jacobson [9] route, with every element given a concrete mechanical realisation in the superfluid.

Step 1: Local Rindler horizon. Consider a defect undergoing uniform acceleration a through the condensate. In the instantaneous rest frame there is a local Rindler horizon: the surface beyond which acoustic signals cannot reach the accelerating observer within a finite proper time. In SSV this horizon is acoustic, located at the locus where the background flow velocity equals the local sound speed c_s.

Step 2: Unruh temperature from the condensate. An accelerating observer perceives a thermal phonon bath at the Unruh temperature [12]

$$T_U = \frac{\hbar\, a}{2\pi\, c\, k_B}. \tag{11}$$

In SSV this is not a formal result about abstract quantum fields: it is the phonon bath of the condensate as measured in an accelerating frame.

Step 3: Entropy proportional to horizon area. The entropy associated with the local horizon is

$$S = \frac{k_B A}{4\xi^2}, \tag{12}$$

where A is the horizon area and $\xi = \hbar/(m_0 c)$ is the healing length. This is the Bekenstein–Hawking area law, but here it is not a postulate: it counts the number of independent vortex-core degrees of freedom that fit on the horizon surface, one per area element ξ^2. The same healing length that fixes the vortex-core size in Paper I simultaneously regulates gravitational entropy.

Step 4: Clausius relation. The heat flux crossing the local horizon is

$$\delta Q = \int_{\mathcal{H}} T_{\mu\nu}\, k^\mu\, d\Sigma^\nu, \tag{13}$$

where k^μ is the horizon-generating Killing vector. Imposing local thermodynamic equilibrium, $\delta Q = T_U\, \delta S$, and substituting (11) and (12):

$$\int_{\mathcal{H}} T_{\mu\nu}\, k^\mu\, d\Sigma^\nu = \frac{\hbar a}{2\pi c k_B} \cdot \frac{k_B}{4\xi^2}\, \delta A. \tag{14}$$

Step 5: Einstein equations. Jacobson [9] showed that requiring (14) to hold for every local Rindler horizon at every spacetime point is equivalent to the Einstein field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}. \tag{15}$$

Newton's constant is fixed by matching the entropy density to the Bekenstein–Hawking formula:

$$G = \frac{c^3 \xi^2}{\hbar} = \frac{c\xi}{m_0}, \tag{16}$$

using $\xi = \hbar/(m_0 c)$. Given the healing length ξ and the fundamental mass m_0, G is determined. Inserting $\xi \sim \ell_P \approx 1.6 \times 10^{-35}$ m and $m_0 \sim m_P$ reproduces $G \approx 6.67 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$.

What SSV adds beyond Jacobson. Jacobson's original derivation requires specifying the entropy density as an external input and does not fix G from first principles. In SSV the UV cutoff ξ is not a free parameter: it is the same healing length that sets the vortex core size, fixes the particle mass ladder (Paper I), and enters the gravitational entropy. The Jacobson argument therefore closes into a genuine derivation of (15) with G expressed in terms of quantities already determined by the particle sector.

Open: quantitative G from $\{m_e, \alpha, \hbar, c\}$

Equation (16) gives the correct order of magnitude but requires $m_0 = m_P$. The Planck mass is not the electron or proton mass. The precise identification of the medium grain mass m_0 with the observed particle masses—and the resulting derivation of G from $\{m_e, \alpha, \hbar, c\}$—is the remaining open problem. It is the same problem as deriving the gravitational fine-structure constant α_G from the proton breather geometry of Paper II, viewed from the thermodynamic direction.

5.2 Cosmological constant

The cosmological constant in (15) corresponds to the residual pressure of the saturated vacuum:

$$\Lambda = \frac{8\pi G}{c^2} \frac{P_0}{\rho_0 c^2}. \tag{17}$$

The cosmological constant problem—why the observed $\Lambda \sim 10^{-52}$ m^{-2} is so much smaller than quantum field theory estimates of vacuum energy—dissolves in SSV. Λ is not a vacuum energy density in the QFT sense but a property of the classical ground state of the superfluid, set by the saturation pressure P_0. The saturation condition suppresses P_0 relative to the naïve zero-point energy, without fine-tuning.

5.3 Planck scale as the healing length

The Planck length is the scale at which the continuum approximation breaks down:

$$\ell_P = \xi = \frac{\hbar}{\sqrt{2m_0 g \rho_0}} \approx 1.6 \times 10^{-35} \text{ m}, \tag{18}$$

where g is the LogSE coupling constant. This fixes $g\rho_0 \sim m_0 c^2/\ell_P^2$. Below ξ the medium is discrete—quantised vortex lines are the fundamental objects, as established in Paper I [1]. Above ξ the continuum description holds and both quantum mechanics and general relativity emerge in the appropriate limits.

Quantum gravity dissolved

The SSV quantum-gravity problem is: derive the continuum Madelung limit from the discrete vortex medium. This is a well-posed coarse-graining problem, not a fundamental incompatibility. The Planck scale is the grain size of the medium, not a new physics threshold. Both quantum mechanics and general relativity are effective descriptions at scales $\gg \xi$.

6 Discussion

6.1 Position in the series

Paper VII-b is the geometry and GR-limit paper in the SSV series. Its inputs are the density-to-potential identification of Paper IV [4], the vortex-core entropy of Paper I [1], the condensate black-hole picture of Paper V [5], and the Madelung equations of Paper VII-a [6]. Its output—the emergent metric, the resolved factor-of-two correction, the Jacobson derivation of Einstein's equations with G in terms of ξ, and the cosmological-constant identification—is the geometrodynamic foundation of the series. Claims about cosmogony and the large-scale structure of spacetime are deferred to Paper VIII.

6.2 Relation to analogue-gravity programme

The Jacobson route used in Section 5.1 and the acoustic-horizon picture of Paper V are closely related to the analogue-gravity programme (Unruh [12, 13], Visser [14], Volovik [15]). The SSV difference is ontological rather than formal: analogue-gravity systems are laboratory analogues of a geometry that exists independently; in SSV the condensate *is* the physical vacuum and the analogy is the fundamental description.

6.3 Open problems

1. *Quantitative G from the particle sector.* Equation (16) gives the correct order of magnitude; deriving G from $\{m_e, \alpha, \hbar, c\}$ without treating $m_0 = m_P$ as an assumption is the principal open problem.

2. *Full Kerr exterior and no-hair theorem.* These are direct consequences of §5.1 once the SSV picture is accepted. The Jacobson route closes onto the *full* nonlinear Einstein equations (15), not specifically the Schwarzschild sector; the Kerr metric is then the unique stationary axisymmetric vacuum solution of those same equations, and the no-hair theorem is the standard GR result of Israel, Carter, Hawking, and Robinson applied to the SSV-emergent metric. Paper V [5] already establishes that black-hole rotation is consistent with the SSV strand-vortex picture via Dirac's trick on the horizon weave, so the rotating condensate black hole is not an obstruction. The remaining work is verification rather than new derivation: explicit construction of the SSV emergent Boyer–Lindquist metric, and a Paper V-style discussion of how the SSV horizon weave realises the Kerr horizon and its frame-dragging.

3. *Strong-field deviations.* The post-Newtonian expansion confirms Einstein's equations perturbatively; the Jacobson argument is nonlinear but assumes local thermodynamic equilibrium. Showing that the condensate dynamics reproduces GR in genuinely strong-field settings (near the horizon at $r \lesssim 10\, r_S$) requires a numerical analysis beyond the present scope.

4. *Dark energy and the late-time Universe.* The cosmological constant identification (17) is qualitative. Deriving the numerical value of Λ from the SSV equation of state and connecting it to the large-scale expansion history is deferred to Paper VIII.

6.4 Success criteria for this paper

The dissolution argument of Section 2 is conceptual; the rest of the paper is the technical content that prevents it from being merely rhetorical. This paper succeeds, on its own terms, if the following are established *within* the scope it claims (post-Newtonian geometry, the Jacobson route, and the dissolution thesis):

1. **The emergent metric is derived, not postulated.** Equation (4) follows from the density-to-potential identification of Paper IV and the Madelung equations of Paper VII-a, with no independent geometric input. This criterion is met for the isotropic post-Newtonian metric and for the time-dilation factor matching the Schwarzschild line element at order Φ/c^2.

2. **The factor-of-two horizon discrepancy of Papers IV and V is resolved by the same emergent-metric machinery.** Including the spatial component $h_{ij} = h_{00}\delta_{ij}$ converts the Newtonian-limit horizon radius $r_H^{(\mathrm{Newt})} = GM/c^2$ into $r_S = 2GM/c^2$, and propagates correctly into $\kappa = c^4/(4GM)$ and the standard Bekenstein–Hawking $T_H = \hbar c^3/(8\pi G M k_B)$ (Section 4.3). This is met in the present paper.

3. **The full Einstein equations are recovered nonlinearly.** The Jacobson five-step route (Section 5.1) closes from a local Rindler horizon and the entropy density $S = k_B A/(4\xi^2)$ to $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$, with the UV cutoff ξ supplied by the particle sector rather than postulated. This is met up to the open problem of fixing m_0 in the entropy formula; the order-of-magnitude G is in place, the quantitative G from $\{m_e, \alpha, \hbar, c\}$ is not.

4. **The dissolution thesis is precise rather than rhetorical.** The replacement of "how do you quantise $g_{\mu\nu}$?" with "coarse-grain the Madelung dynamics over scales $\gg \xi$" is a definite mathematical programme: the inputs are the LogSE equation of state and the vortex spectrum of Paper I; the output is the metric of Equation (4); the missing step is the coarse-graining rigorously connecting them. This paper meets this criterion at the level of identifying the programme and ruling out the alternative (quantising the metric as an independent field).

5. **No claims are made outside the paper's stated scope.** Black-hole interior ontology stays in Paper V; the Madelung derivation of quantum mechanics stays in Paper VII-a; cosmogony and the numerical value of Λ stay in Paper VIII; and the strong-field validation, quantitative G, and Kerr exterior are explicitly listed as open problems above rather than implicitly claimed.

What would constitute closure-grade success. Each open problem in Section 6.3 has a definite upgrade condition: a derivation of G from $\{m_e, \alpha, \hbar, c\}$ that does not require $m_0 = m_P$ as an input (problem 1); a Kerr-metric extension of the Jacobson argument and an acoustic-horizon analogue of the no-hair theorem (problem 2); a numerical demonstration that the condensate dynamics reproduces GR in genuinely strong-field settings, e.g. near $r \sim r_S$ (problem 3); and a quantitative Λ predicted from the saturation equation of state together with a connection to the observed expansion history (problem 4). Each is a separate workstream that would, when complete, lift one of the above criteria from the present-paper level to closure-grade.

7 Conclusion

If geometry is emergent from the condensate, the quantum-gravity problem is replaced by a derivation problem. The central task is not to quantise spacetime but to show that spacetime-like behaviour coarse-grains from the vortex dynamics of the saturated vacuum. The results of this paper are:

1. The emergent metric (4) follows directly from the density-to-potential identification of Paper IV, with no separate geometric postulate.

2. Spatial curvature in the isotropic metric resolves the factor-of-two horizon discrepancy of Papers IV and V, yielding the correct Schwarzschild radius $r_H = 2GM/c^2$, surface gravity (9), and Hawking temperature (10).

3. The full nonlinear Einstein equations (15) follow from the Jacobson acoustic-thermodynamic route, with G expressed in terms of the healing length ξ already fixed by the particle sector.

4. The cosmological constant (17) is the saturation pressure of the vacuum ground state, not a zero-point energy density; no fine-tuning is required.

5. The Planck scale is the vortex-core healing length: the scale at which the continuum description breaks down. Quantum mechanics and general relativity are both effective theories at scales $\gg \xi$.

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